

Assignment

Marks 30

1. Describe the role of sensors, actuators, and electromechanical devices in feedback control systems.
2. Discuss why servomotors are preferred over conventional motors in control systems.
3. Justify the use of optical encoders in precision positioning systems.
4. Evaluate the selection criteria for choosing a position sensor in an industrial control system.
5. Explain the importance of tachometer feedback in improving system stability and performance.
6. Explain why stepper motors are widely used in CNC machines and robotic systems.